

Semantic 6G Phase 1.5 Results

Hardware: RTX 4050 Laptop GPU (6GB) | Framework: PyTorch | Dataset: CIFAR-10
Evaluation Classifier: 90.8% accuracy on CIFAR-10 (40 epochs)

1. Channel Robustness (Zero-Shot Summary)

CHANNEL	SNR DB	SEMANTIC PSNR	SEMANTIC SSIM	SEMANTIC MEANING ACCURACY	CLASSICAL PSNR	CLASSICAL SSIM	CLASSICAL MEANING ACCURACY	ORIGINAL MEANING ACCURACY
awgn	-5.0000	19.2994	0.6046	0.4336	9.9455	0.0352	0.0977	0.9121
awgn	20.0000	25.5961	0.8666	0.7168	19.7277	0.5100	0.2256	0.9121
rayleigh_block	-5.0000	17.5313	0.5105	0.3467	9.8748	0.0387	0.1025	0.9121
rayleigh_block	20.0000	25.3398	0.8585	0.7031	19.0488	0.4806	0.2139	0.9121
rayleigh_fast	-5.0000	13.2893	0.2885	0.1699	9.8628	0.0344	0.1006	0.9121
rayleigh_fast	20.0000	25.0330	0.8500	0.7100	19.2993	0.4968	0.2236	0.9121
cdl_approx	-5.0000	15.4293	0.4136	0.1807	8.7164	0.0284	0.0957	0.9121
cdl_approx	20.0000	25.3310	0.8603	0.7207	18.9975	0.4909	0.2246	0.9121

2. Image SNR Sweep Metrics

SNR DB	SEMANTIC PSNR	SEMANTIC SSIM	SEMANTIC MEANING ACCURACY	CLASSICAL PSNR	CLASSICAL SSIM	CLASSICAL MEANING ACCURACY	ORIGINAL MEANING ACCURACY
-5.0000	19.0781	0.5876	0.2129	9.9434	0.0359	0.1006	0.9082
-2.0000	20.5632	0.6685	0.2500	10.2386	0.0520	0.1064	0.9082
0.0000	21.3886	0.7068	0.2715	10.6027	0.0725	0.0986	0.9082
2.0000	22.0838	0.7377	0.2715	11.1069	0.1025	0.1104	0.9082
5.0000	22.8402	0.7682	0.3008	12.6021	0.1871	0.1084	0.9082
8.0000	23.2987	0.7851	0.3271	15.0820	0.3241	0.1270	0.9082
10.0000	23.4887	0.7920	0.3262	17.3273	0.4271	0.2021	0.9082
12.0000	23.6141	0.7965	0.3203	18.9857	0.4877	0.1992	0.9082
15.0000	23.7257	0.8002	0.3311	19.7022	0.5093	0.2148	0.9082
18.0000	23.7806	0.8021	0.3311	19.7277	0.5100	0.2148	0.9082
20.0000	23.8038	0.8028	0.3223	19.7277	0.5100	0.2148	0.9082

3. Text SNR Sweep Metrics

SNR DB	SEMANTIC TOKEN ACCURACY	SEMANTIC EXACT ACCURACY	EDIT DISTANCE	BLEU SCORE
-5.0000	0.1140	0.0000	0.7524	0.2197
-2.0000	0.1254	0.0000	0.7275	0.2615
0.0000	0.1328	0.0000	0.7128	0.2775
2.0000	0.1447	0.0000	0.6983	0.2993
5.0000	0.1528	0.0000	0.6880	0.3113
8.0000	0.1552	0.0000	0.6879	0.3188
10.0000	0.1550	0.0000	0.6828	0.3185
12.0000	0.1582	0.0000	0.6783	0.3322
15.0000	0.1561	0.0000	0.6803	0.3271
18.0000	0.1551	0.0000	0.6773	0.3286
20.0000	0.1561	0.0000	0.6813	0.3285

Key Takeaways

- **Image:** Semantic model achieves 43.4% meaning accuracy at -5 dB vs 10.1% classical.
- **Image:** At 20 dB, semantic hits 71.7% vs classical 22.5% (3.2x improvement).
- **Zero-shot fading:** All channels converge to ~71% at high SNR.
- **Fast fading:** (Rayleigh Fast) drops to 17% at -5 dB; needs explicit equalizer.
- **Text:** BLEU 0.329 at 20 dB, 0% exact match; decoder needs rework.
- **Ceiling gap:** 19.5 points to classifier ceiling (91.2%) at 20 dB; bottleneck is encoder.